Table 1: **Supplementary Table 5: Statistical comparison and model selection of constrained Gaussian mixture models on mRNA–protein fold changes.** Systematic evaluation of bivariate Gaussian mixture models across cohorts (BDL and CCl_4) and elongation factors $c \in \{1.3, 1.5, 2.0, 2.5, 4.0\}$. Models were fitted using an Expectation-Maximization (EM) algorithm to joint distributions of $N = 943$ matched gene–protein pairs. Number of free parameters P, Log-Likelihood ($\log L$), Bayesian Information Criterion (BIC), $\Delta\text{BIC} = \text{BIC} - \text{BIC}_{K=1}$, and estimated mixing proportions π_k (Diagonal π_1, Vertical π_2, Horizontal π_3). Highlighted rows denote the overall winning model for each etiology (lowest BIC).

Cohort	Model	c	Ratio $\sqrt{c}$	P	$\log L$	BIC	ΔBIC	π_1 (Diag)	π_2 (Vert)	π_3 (Horiz)	Status
Cholestatic Liver Fibrosis (BDL, $N = 943$ pairs)											
BDL	$K = 1$	–	–	5	-1830.5	3695.3	+0.0	100.0%	–	–	Null Single Gaussian
BDL	$K = 2$	1.3	1.14	10	-1661.0	**3390.5**	**-304.8**	90.9%	9.1%	–	**Winning Model (Lowest BIC)**
BDL	$K = 2$	1.5	1.22	10	-1662.3	3393.0	-302.2	91.1%	8.9%	–	Constrained Model
BDL	$K = 2$	2.0	1.41	10	-1666.8	3402.0	-293.3	91.5%	8.5%	–	Constrained Model
BDL	$K = 2$	2.5	1.58	10	-1671.5	3411.5	-283.7	92.0%	8.0%	–	Constrained Model
BDL	$K = 2$	4.0	2.00	10	-1683.5	3435.6	-259.7	92.8%	7.2%	–	Constrained Model
BDL	$K = 3$	1.3	1.14	15	-1647.9	3398.5	-296.8	85.8%	8.8%	5.4%	Constrained Model
BDL	$K = 3$	1.5	1.22	15	-1649.4	3401.5	-293.7	86.0%	8.8%	5.1%	Constrained Model
BDL	$K = 3$	2.0	1.41	15	-1653.2	3409.1	-286.2	86.5%	8.9%	4.6%	Constrained Model
BDL	$K = 3$	2.5	1.58	15	-1656.4	3415.6	-279.7	86.8%	9.0%	4.3%	Constrained Model
BDL	$K = 3$	4.0	2.00	15	-1663.4	3429.5	-265.8	87.4%	9.0%	3.7%	Constrained Model
Toxic Liver Fibrosis (CCl_4, $N = 943$ pairs)											
CCl4	$K = 1$	–	–	5	-1839.3	3712.9	+0.0	100.0%	–	–	Null Single Gaussian
CCl4	$K = 2$	1.3	1.14	10	-1768.9	3606.4	-106.5	69.5%	30.5%	–	Constrained Model
CCl4	$K = 2$	1.5	1.22	10	-1768.9	3606.4	-106.5	69.5%	30.5%	–	Constrained Model
CCl4	$K = 2$	2.0	1.41	10	-1768.9	3606.4	-106.5	69.5%	30.5%	–	Constrained Model
CCl4	$K = 2$	2.5	1.58	10	-1768.9	3606.4	-106.5	69.5%	30.5%	–	Constrained Model
CCl4	$K = 2$	4.0	2.00	10	-1751.5	3571.6	-141.3	93.7%	6.3%	–	Constrained Model
CCl4	$K = 3$	1.3	1.14	15	-1693.9	**3490.6**	**-222.3**	61.6%	30.2%	8.2%	**Winning Model (Lowest BIC)**
CCl4	$K = 3$	1.5	1.22	15	-1696.0	3494.7	-218.2	62.9%	30.6%	6.5%	Constrained Model
CCl4	$K = 3$	2.0	1.41	15	-1700.2	3503.2	-209.7	64.4%	30.7%	4.8%	Constrained Model
CCl4	$K = 3$	2.5	1.58	15	-1703.5	3509.7	-203.2	65.3%	30.7%	4.0%	Constrained Model
CCl4	$K = 3$	4.0	2.00	15	-1709.8	3522.3	-190.6	66.4%	30.8%	2.8%	Constrained Model